EXPERIMENT - 1

AIM: To Implement Linear and Logistic Regression on real-world dataset.- [expanded_student_performance.csv]

OUTPUT :

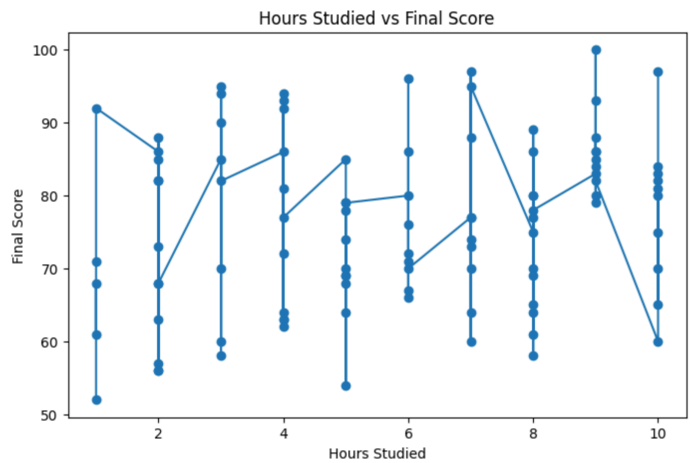
Loading the dataset

```
First 5 rows:
   Hours_Studied  Attendance  Assignment_Score  Midterm_Score  Final_Score
0              1        60.0              55.0           50.0           52
1              2        65.0              58.0           55.0           57
2              3        70.0              60.0           58.0           60
3              4        75.0              65.0           62.0           64
4              5        80.0              68.0           65.0           68

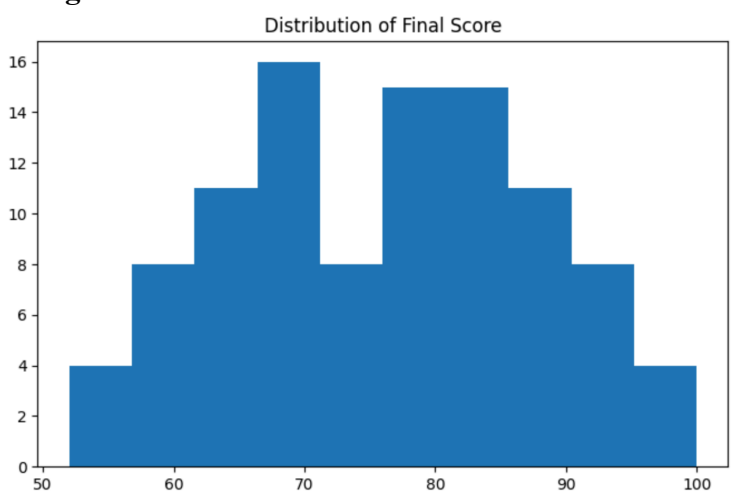
Dataset Shape: (100, 5)

Missing Values:
 Hours_Studied        0
Attendance          11
Assignment_Score    12
Midterm_Score       19
Final_Score          0
dtype: int64
```

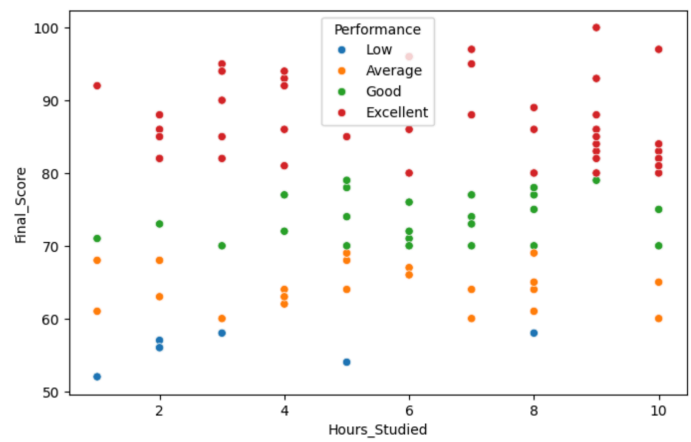
Line plot



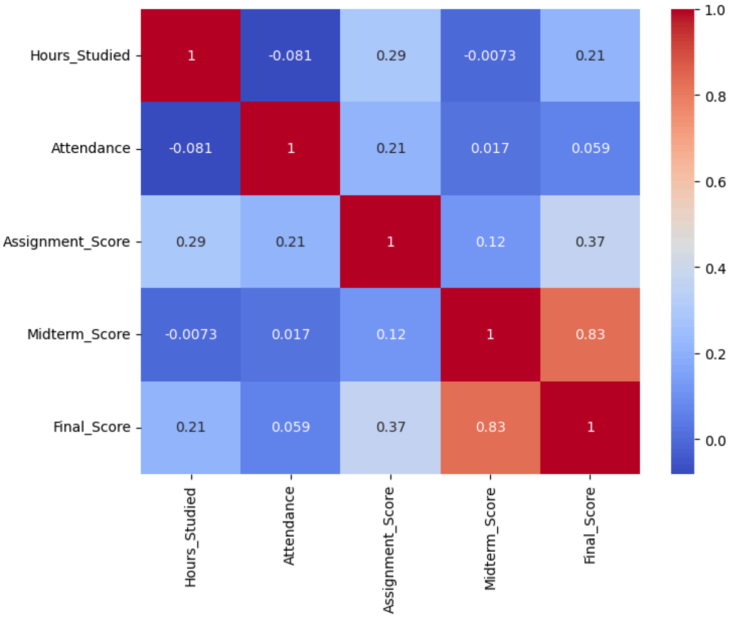
Histogram



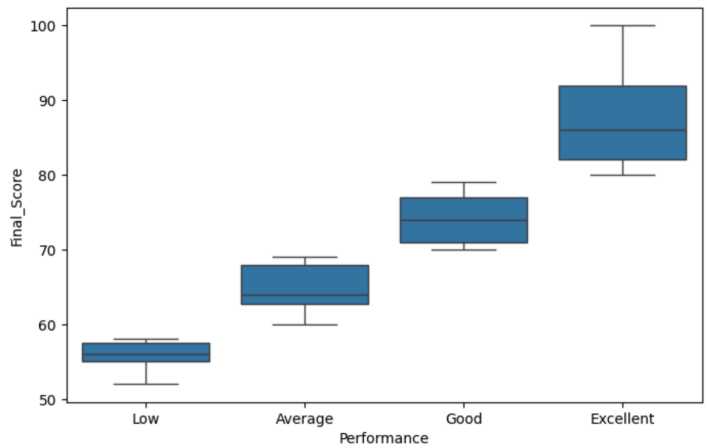
Scatter Plot



Heatmap



Boxplot



Linear regression MSE AND R2 score

```
MSE: 35.62314724702965
R2 Score: 0.7102133797107029
```

Logistic regression

```
Accuracy: 0.8666666666666667

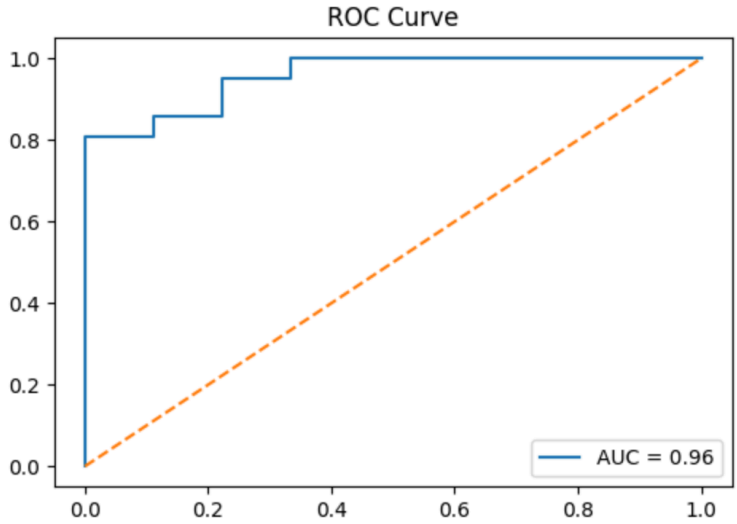
Confusion Matrix:
 [[ 6  3]
 [ 1 20]]

Classification Report:
              precision    recall  f1-score   support

           0       0.86      0.67      0.75         9
           1       0.87      0.95      0.91        21

    accuracy                           0.87        30
   macro avg       0.86      0.81      0.83        30
weighted avg       0.87      0.87      0.86        30
```

ROC Curve



Hyperparameter tuning

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Best Parameters: {'C': 10, 'solver': 'liblinear'}
Best Score: 0.8428571428571429
```

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